

1. The Advection-Diffusion Equation

1.1	Derivation of the advection-diffusion equation	3
1.2	The 1D advection equation	6
1.3	The 1D diffusion equation	8
1.4	The 1D advection-diffusion equation	9
1.5	The Péclet number	10
1.6	The Burgers' equation	11
1.7	Summary of key concepts	13

Introduction

This course focuses on the numerical simulation of fluid flow and heat transport, which are governed by the following conservation equations for mass, momentum, and energy, assuming a Newtonian fluid,

$$\begin{aligned}
 \text{mass:} \quad & \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 & (1.1a) \\
 \text{momentum:} \quad & \frac{\partial}{\partial t}(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = \nabla \cdot (\mu \nabla \mathbf{u}) - \nabla p + \nabla \cdot \mathbf{B} & (1.1b) \\
 \text{energy:} \quad & \frac{\partial}{\partial t}(\rho e) + \nabla \cdot (\rho e \mathbf{u}) = \nabla \cdot (k \nabla T) + \boldsymbol{\sigma} : \nabla \mathbf{u} & (1.1c)
 \end{aligned}$$

unsteady
advection
diffusion

where ρ , μ , k , $\mathbf{u}$, p , e , and $\boldsymbol{\sigma}$ denote the fluid density, viscosity, thermal conductivity, velocity, pressure, specific internal energy, and stress tensor, respectively. The $\mathbf{B}$ term in the momentum equation (1.1b) accounts for additional viscous stresses arising due to variations in density and viscosity,

$$\mathbf{B} = \mu(\nabla \mathbf{u}^T) - \frac{2}{3}\mu(\nabla \cdot \mathbf{u})\mathbf{I}. \quad (1.2)$$

Despite describing different physical quantities, the three conservation equations in (1.1) share some common structures. For example, the left-hand side of each equation can be written as

$$\underbrace{\frac{\partial}{\partial t}(\rho \phi)}_{\text{unsteady}} + \underbrace{\nabla \cdot (\rho \mathbf{u} \phi)}_{\text{advection}}, \quad (1.3)$$

where $\phi = 1$, $\mathbf{u}$, or e . The first term above represents the temporal accumulation of $\rho \phi$ within a control volume, and is often called the “*unsteady*” or “*accumulation*” term. The second term represents the transport of ϕ by the flow, and is called the “*advection*” or “*convection*” term.¹

¹See *Side Discussion* (1.1) for a brief note about our definitions of the terms *advection* and *convection*.

If we turn our attention to the right-hand-sides of the three conservation equations in (1.1), we see that the momentum and energy equations include “*diffusion*” terms of the form

$$\underbrace{\nabla \cdot (\Gamma \nabla \phi)}_{\text{diffusion}}, \quad (1.4)$$

which model the molecular diffusion of ϕ . For momentum, $(\Gamma, \phi) = (\mu, \mathbf{u})$, and for energy, $(\Gamma, \phi) = (k, T)$.

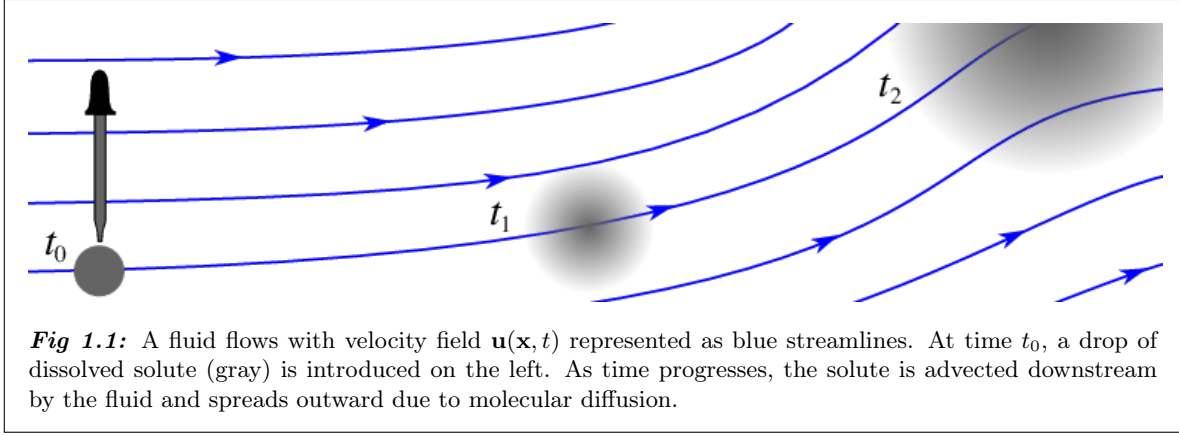
The interplay between unsteady, advective, and diffusive terms is a core issue in CFD, because it strongly influences the stability and accuracy of simulations. The initial chapters of this textbook examine these issues using the following advection–diffusion equation for a passive scalar,

$$\underbrace{\frac{\partial}{\partial t}(\rho\phi)}_{\text{unsteady}} + \underbrace{\nabla \cdot (\rho \mathbf{u} \phi)}_{\text{advection}} = \underbrace{\nabla \cdot (\Gamma \nabla \phi)}_{\text{diffusion}} + \underbrace{f(\mathbf{x}, t)}_{\text{source}}. \quad (1.5)$$

We begin in Section 1.1 by deriving Eq. (1.5). We then explore the roles of unsteadiness, advection, and diffusion analytically.

Side Note 1.1: Advection vs. convection

The terms *advection* and *convection* are used inconsistently across scientific communities, and their meanings have evolved over time. As a result, Eq. (1.5) is also often called the *convection–diffusion* equation. In this text, we adopt the convention that *advection* refers to transport by the fluid velocity field $\mathbf{u}$, while *convection* refers more broadly to combined advective and diffusive transport. This is a practical choice made for clarity and consistency, and we do not claim it to be the only valid definition.



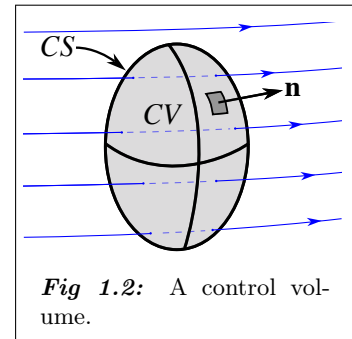
1.1 Derivation of the advection-diffusion equation

Here we derive the advection–diffusion equation governing the transport of a *passive scalar* in a fluid flow. A passive scalar is a scalar field, such as concentration, that is transported within a fluid without affecting the fluid’s motion. Common examples include dissolved solutes, such as sugar or salt, provided they do not significantly alter the fluid’s thermophysical properties, such as density or viscosity. To motivate our derivation, consider the situation sketched in Fig. 7.3, where a fluid flows with velocity field $\mathbf{u}(\mathbf{x}, t)$. At time t_0 , a solute is introduced on the left and subsequently transported rightward by *advection* and *diffusion*. Advection refers to the downstream transport of solute by the fluid motion, while diffusion describes its tendency to spread from regions of high to low solute concentration.

1.1.1 Control volume mass balance

To derive an equation governing the solute transport, we introduce a fixed control volume CV in the fluid, as in Fig. 1.2. The volume is bounded by a control surface CS with outward unit normal $\mathbf{n}$. We assume the solute mass $m(t)$ in CV can change due to three mechanisms:

1. transport across CS by advection,
2. transport across CS by diffusion,
3. generation or consumption within CV due to phenomena such as chemical reactions.



We express this balance as

$$\frac{dm}{dt} = -\dot{m}_{\text{adv}} - \dot{m}_{\text{diff}} + \dot{m}_{\text{src}}, \quad (1.6)$$

where $\dot{m}_{\text{adv}}$ and $\dot{m}_{\text{diff}}$ are the rates (kg/s) at which solute leaves the control volume by advection and diffusion, and $\dot{m}_{\text{src}}$ is the net rate of solute generation in the volume CV .

1.1.2 Measuring concentration

At this stage, we must choose a representation for solute concentration. One common measure is the *mass concentration* c , defined as

$$c = \frac{\text{solute mass}}{\text{volume of solution}}. \quad (1.7)$$

However, in finite-volume methods, it is often more convenient to use the *mass fraction* ϕ , defined as

$$\phi = \frac{\text{solute mass}}{\text{solution mass}} = \frac{c}{\rho}, \quad (1.8)$$

where ρ is the solution density. In terms of ϕ , the solute mass in a control volume is

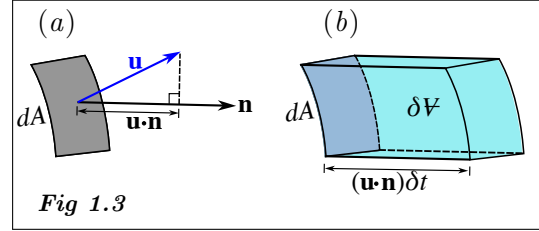
$$m(t) = \int_{CV} \rho \phi dV, \quad (1.9)$$

where ρ and ϕ vary with $\mathbf{x}$ and t .

To be consistent with classic textbooks [1], we primarily represent concentration using ϕ . However, depending on the context, we occasionally also use c .

1.1.3 Advective mass transport

To continue the derivation, we express $\dot{m}_{\text{adv}}$ and $\dot{m}_{\text{diff}}$ in terms of ρ and ϕ . For advection, we consider the differential element dA of the control surface sketched in Fig. 1.3(a). Over a short time interval δt , fluid travels a distance $(\mathbf{u} \cdot \mathbf{n}) \delta t$ normal to the surface. The corresponding fluid volume crossing dA is,



$$\delta V = (\mathbf{u} \cdot \mathbf{n}) \delta t dA, \quad (1.10)$$

where δV is positive for outflow and negative for inflow. The corresponding solute mass crossing the surface element is

$$\delta m = \rho \phi \delta V = \rho \phi (\mathbf{u} \cdot \mathbf{n}) \delta t dA. \quad (1.11)$$

Dividing this expression by δt gives the corresponding mass flux rate,

$$\frac{\delta m}{\delta t} = \rho \phi (\mathbf{u} \cdot \mathbf{n}) dA. \quad (1.12)$$

Integrating this expression over the control surface gives the following expression for $\dot{m}_{\text{adv}}$,

$$\dot{m}_{\text{adv}} = \int_{CS} \rho \phi (\mathbf{u} \cdot \mathbf{n}) dA. \quad (1.13)$$

1.1.4 Diffusive mass transport

To express $\dot{m}_{\text{diff}}$ in terms of ρ and ϕ , we use Fick's law of mass diffusion,

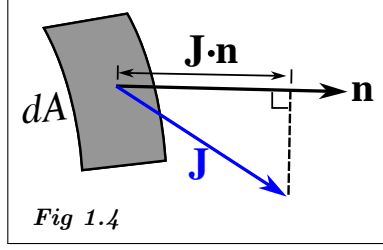
$$\mathbf{J} = -\rho \mathcal{D} \nabla \phi, \quad (1.14)$$

where $\mathbf{J}$ is the diffusive mass flux ($\text{kg m}^{-2} \text{s}^{-1}$), and $\mathcal{D}$ is the mass diffusivity ($\text{m}^2 \text{s}^{-1}$). The negative sign indicates that passive scalars diffuse from high to low concentration. For a differential surface element dA (Fig. 1.4), the diffusive mass flux across dA is

$$\delta \dot{m} = \mathbf{J} \cdot \mathbf{n} dA = -(\rho \mathcal{D} \nabla \phi) \cdot \mathbf{n} dA. \quad (1.15)$$

Integrating this expression over the control surface gives the following expression for $\dot{m}_{\text{diff}}$,

$$\dot{m}_{\text{diff}} = - \int_{CS} (\rho \mathcal{D} \nabla \phi) \cdot \mathbf{n} dA. \quad (1.16)$$



1.1.5 Solute generation

Finally, we express the source term as the following volume integral,

$$\dot{m}_{\text{src}} = \int_{CV} f(\mathbf{x}, t) dV, \quad (1.17)$$

where $f(\mathbf{x}, t)$ is the solute generation rate per unit volume ($\text{kg m}^{-3} \text{s}^{-1}$).

1.1.6 Integral and differential forms

Combining our expressions for $\dot{m}_{\text{adv}}$, $\dot{m}_{\text{diff}}$, and $\dot{m}_{\text{src}}$ produces the following integral form of the advection-diffusion equation,

$$\boxed{\frac{d}{dt} \int_{CV} \rho \phi dV + \int_{CS} (\rho \mathbf{u} \phi) \cdot \mathbf{n} dA = \int_{CS} (\Gamma \nabla \phi) \cdot \mathbf{n} dA + \int_{CV} f(\mathbf{x}, t) dV,} \quad (1.18)$$

with $\Gamma = \rho \mathcal{D}$. Equation (1.18) is the integral form of the advection-diffusion equation used in finite-volume methods. However, when seeking analytical solutions, we typically convert it to a differential equation. For that, we bring the time derivative inside the volume integral and apply Gauss' theorem to express the advection-diffusion equation as a single volume integral

$$\int_{CV} \left[\frac{\partial}{\partial t} (\rho \phi) + \nabla \cdot (\rho \mathbf{u} \phi) - \nabla \cdot (\Gamma \nabla \phi) - f(\mathbf{x}, t) \right] dV = 0. \quad (1.19)$$

Because CV is arbitrary, the integrand above must vanish, yielding the differential equation

$$\boxed{\frac{\partial}{\partial t} (\rho \phi) + \nabla \cdot (\rho \mathbf{u} \phi) = \nabla \cdot (\Gamma \nabla \phi) + f(\mathbf{x}, t).} \quad (1.20)$$

Although we derived Eq. (1.20) in the context of solute transport, the same equation governs the transport of any passive scalar ϕ , provided we interpret Γ as the appropriate diffusivity for that quantity.

For the scope of the remaining sections in this chapter, we make the following assumptions that allow us to explore the advection–diffusion equation analytically:

1. ρ and Γ are constant,
2. the velocity field is produced by an incompressible fluid flow, so that $(\nabla \cdot \mathbf{u} = 0)$,
3. the source term is zero, $f(\mathbf{x}, t) = 0$.

Under these assumptions, the advection–diffusion equation simplifies to

$$\rho \left(\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi \right) = \Gamma \nabla^2 \phi. \quad (1.21)$$

1.2 The 1D advection equation

When diffusion is negligible, the advection–diffusion equation (1.21) reduces to the *advection equation* below,

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = 0. \quad (1.22)$$

This can be written more succinctly as,

$$\frac{D\phi}{Dt} = 0, \quad (1.23)$$

where D/Dt is the material derivative,

$$\frac{D(\cdot)}{Dt} = \frac{\partial(\cdot)}{\partial t} + \mathbf{u} \cdot \nabla(\cdot). \quad (1.24)$$

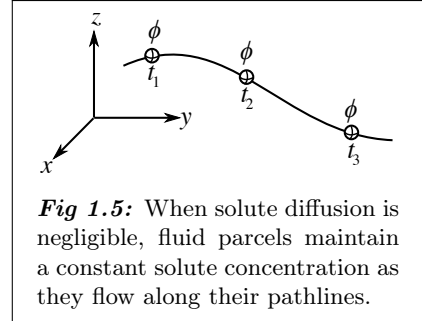
From the definition of the material derivative,² Eq. (1.23) implies that as a small fluid parcel travels along its pathline, the mass fraction ϕ of solute in the parcel remains constant, as sketched in Fig. 1.5. This occurs because solutes cannot leave the parcel via diffusion, and because the parcel volume remains constant for an incompressible flow.

To examine the advection equation further, let's consider the one-dimensional version,

$$\frac{\partial \phi}{\partial t} + u \frac{\partial \phi}{\partial x} = 0, \quad (1.25)$$

where u is a constant. Equation (1.25) produces wave-like solutions in which information travels at the speed u . This behavior occurs because Eq. (1.25) is automatically satisfied by any concentration field of the form

$$\phi(x, t) = \phi(x - ut), \quad (1.26)$$

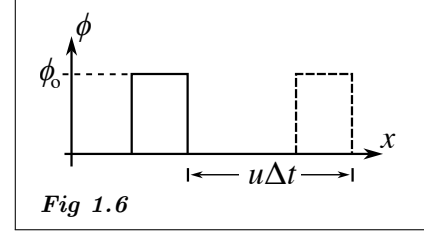


²Recall that the material derivative $D\alpha/Dt$ of some property α represents the time rate-of-change of α in a Lagrangian reference frame.

To see why, define the variable $\xi = x - ut$ and let $\phi(x, t) = \phi(\xi)$. Using the chain rule, we can then show that

$$\frac{\partial \phi}{\partial t} = \frac{d\phi}{d\xi} \frac{\partial \xi}{\partial t} = -u \frac{d\phi}{d\xi}, \quad (1.27)$$

$$\frac{\partial \phi}{\partial x} = \frac{d\phi}{d\xi} \frac{\partial \xi}{\partial x} = \frac{d\phi}{d\xi}. \quad (1.28)$$

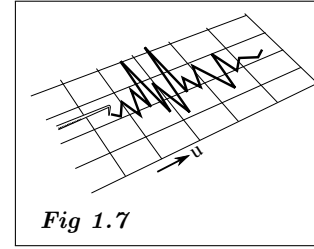


Substituting these relations in Eq. (1.25) confirms that solutions of the form $\phi(\xi)$ satisfy

$$\frac{\partial \phi}{\partial t} + u \frac{\partial \phi}{\partial x} = -u \frac{d\phi}{d\xi} + u \frac{d\phi}{d\xi} = 0. \quad (1.29)$$

Suppose that at the initial time $t = 0$, the one-dimensional advection equation is subject to an initial concentration pulse that is everywhere zero except for a small region in which $\phi = \phi_0$, as in Fig. 1.6(a). If an observer travels rightward with velocity u , the observer's x -position increases as $x = ut$. In the observer's moving reference frame, the variable $\xi = x - ut$ does not vary with time, and the concentration pulse appears fixed. However, in a stationary reference frame, the pulse translates rightward with the constant speed u , as sketched in Fig. 1.6(a).

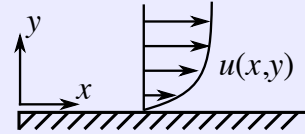
Because the one-dimensional advection equation is first order, it can only satisfy one boundary condition. Because information only travels in the downstream direction, this boundary condition should be applied at the upstream boundary of the domain. In this sense, one can think of solutions to the advection equation as being similar to plots made by a seismograph in which the paper travels rightward with velocity u , and the pen plays the role of the upstream boundary condition, as sketched in Fig. 1.7.



Side Note 1.2: Application to boundary layers

A key example of competing advection and diffusion effects occurs in boundary layers. A steady, incompressible, 2D boundary layer satisfies the momentum equation below,

$$\rho u \frac{\partial u}{\partial x} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \underbrace{\mu \frac{\partial^2 u}{\partial x^2}}_{\text{downstream diffusion}} + \mu \frac{\partial^2 u}{\partial y^2}, \quad (1.30)$$



where x is the downstream flow direction (as sketched). In boundary layers, the downstream diffusion term is typically negligible, and Eq. (1.30) can be reduced to

$$\underbrace{\rho u \frac{\partial u}{\partial x}}_{\text{downstream advection}} + \rho v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2}. \quad (1.31)$$

However, after this simplification, the only x -derivative occurs in the downstream advection term. Because the remaining derivative is only first-order, Eq. (1.31) can satisfy only one boundary condition in the x -direction. This condition must be applied upstream, because advection only transports information downstream.

1.3 The 1D diffusion equation

Next, we consider the case of a solute diffusing in a fluid at rest, $\mathbf{u} = 0$. In this situation, the advection-diffusion equation reduces to the *diffusion equation* below,

$$\frac{\partial \phi}{\partial t} = \mathcal{D} \nabla^2 \phi, \quad (1.32)$$

where we set $\mathcal{D} = \Gamma/\rho$. The diffusion Eq. (1.32) produces smooth solutions in which information propagates in all directions. For steady boundary conditions, the equation typically describes a transient process in which local maxima or minima are smoothed out until the field reaches a steady state for which $\nabla^2 \phi = 0$. In this final state, maxima or minima can only occur on the boundaries.

To demonstrate this behavior, consider the one-dimensional diffusion equation

$$\frac{\partial \phi}{\partial t} = \mathcal{D} \frac{\partial^2 \phi}{\partial x^2}, \quad (1.33)$$

subject to the initial condition

$$\phi(x, 0) = \begin{cases} \phi_o, & |x| \leq 1, \\ 0, & |x| > 1 \end{cases} \quad (1.34)$$

shown as a black line in Fig. 1.8. When solved subject to the boundary conditions $\phi \rightarrow 0$ as $x \rightarrow \pm\infty$, the diffusion equation (1.33) has the exact solution

$$\frac{\phi(x, t)}{\phi_o} = \frac{1}{2} \left[\operatorname{erf} \left(\frac{x+1}{\sqrt{4\mathcal{D}t}} \right) - \operatorname{erf} \left(\frac{x-1}{\sqrt{4\mathcal{D}t}} \right) \right]. \quad (1.35)$$

Figure 1.8 shows this solution at times $t = 0.01, 0.1$, and 1 , with $\mathcal{D} = 1$. We see that as $t \rightarrow \infty$, the concentration field approaches the uniform value $\phi = 0$. Moreover, in contrast to the case of advection, we see that diffusion transports information in both directions ($\pm x$). Mathematically, this occurs because diffusion involves second-order derivatives in space.

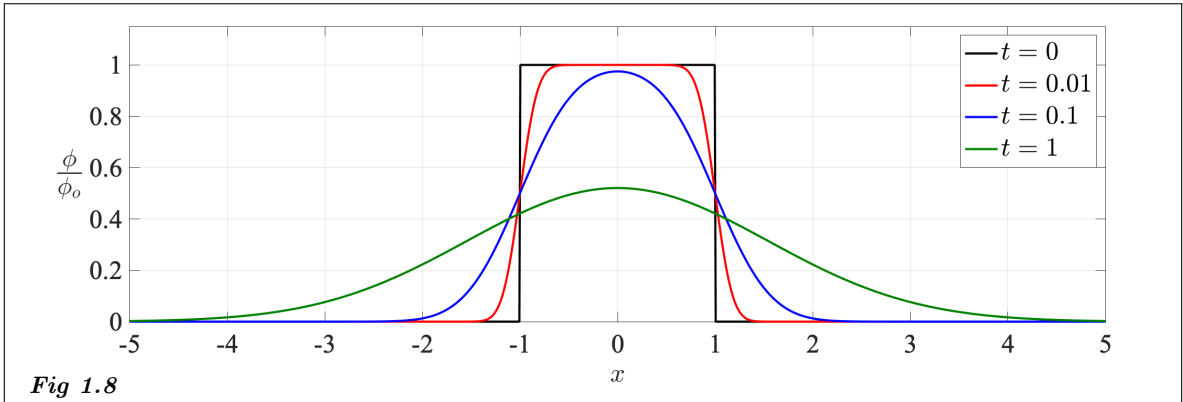
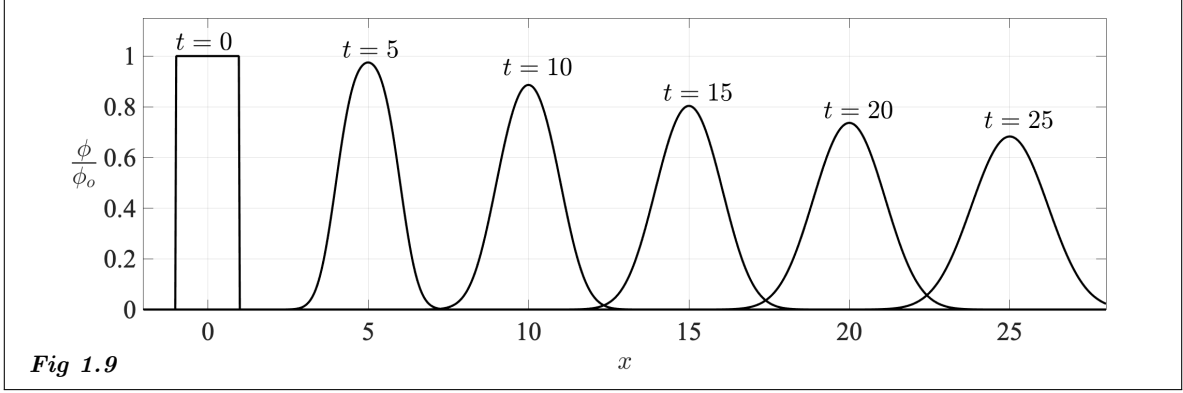


Fig 1.8



1.4 The 1D advection-diffusion equation

Next, we consider the combined effects of advection and diffusion by solving the one-dimensional advection-diffusion equation

$$\frac{\partial \phi}{\partial t} + u \frac{\partial \phi}{\partial x} = \mathcal{D} \frac{\partial^2 \phi}{\partial x^2}, \quad (1.36)$$

where u is constant. We solve Eq. (1.36) subject to the initial condition

$$\phi(x, 0) = \begin{cases} \phi_o, & |x| \leq 1, \\ 0, & |x| > 1 \end{cases} \quad (1.37)$$

and the boundary conditions $\phi \rightarrow 0$ as $x \rightarrow \pm\infty$. In that case, Eq. (1.36) has the exact solution

$$\frac{\phi(x, t)}{\phi_o} = \frac{1}{2} \left[\operatorname{erf} \left(\frac{x - ut + 1}{\sqrt{4\mathcal{D}t}} \right) - \operatorname{erf} \left(\frac{x - ut - 1}{\sqrt{4\mathcal{D}t}} \right) \right]. \quad (1.38)$$

Figure 1.9 shows the subsequent evolution of the pulse when $u = 1$ and $\mathcal{D} = 0.02$. We see that under the combined effects of advection and diffusion, the pulse is transported rightward at the velocity $u = 1$, while being smoothed by diffusion. Figure 1.9 illustrates an important concept called *boundedness*. This principle states that advection and diffusion do not create new maxima or minima, so that the solution at any time remains within the range of the initial condition. If a steady state is reached, the extrema can occur only at the boundaries. Boundedness is important in CFD because the discretization of advection and diffusion terms should preserve this property in the numerical solution.

1.5 The Péclet number

In flows where advection and diffusion occur simultaneously, their relative importance is often characterized by a non-dimensional parameter called the *Péclet number*. For illustration, consider a steady, incompressible flow between two porous plates, as sketched in Fig. 1.10. The plate at $x = 0$ injects fluid rightward with uniform velocity U , while the plate at $x = L$ removes fluid at the same rate. Under these conditions, the incompressible Navier–Stokes and continuity equations yield the uniform velocity field $\mathbf{u} = U\mathbf{i}$.

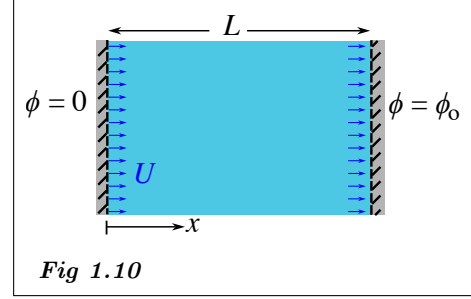


Fig 1.10

Now suppose the solute concentration is fixed to $\phi = 0$ on the left plate, and to $\phi = \phi_o$ on the right. If we introduce the non-dimensional variables $\hat{x} = x/L$ and $\hat{\phi} = \phi/\phi_o$, the steady-state concentration then satisfies the advection–diffusion problem

$$\frac{d\hat{\phi}}{d\hat{x}} = \frac{1}{Pe} \frac{d^2\hat{\phi}}{d\hat{x}^2}, \quad \hat{\phi}|_{\hat{x}=0} = 0, \quad \hat{\phi}|_{\hat{x}=1} = 1, \quad (1.39)$$

where Pe is the *Péclet number*,

$$Pe = \frac{UL}{\mathcal{D}} = \frac{\rho UL}{\Gamma}. \quad (1.40)$$

Equation (1.39) admits the analytical solution

$$\hat{\phi} = \frac{e^{Pe\hat{x}} - 1}{e^{Pe} - 1}. \quad (1.41)$$

The Péclet number in Eqs. (1.39–1.41) can be interpreted as the ratio of a diffusive time scale τ_{diff} to an advective time scale τ_{adv} ,

$$Pe = \frac{\tau_{\text{diff}}}{\tau_{\text{adv}}}, \quad \tau_{\text{adv}} = \frac{L}{U}, \quad \tau_{\text{diff}} = \frac{L^2}{\mathcal{D}}. \quad (1.42)$$

When diffusion acts much faster than advection, the Péclet number is small,

$$\tau_{\text{diff}} \ll \tau_{\text{adv}}, \quad Pe \ll 1. \quad (1.43)$$

This occurs when the velocity is small, $U \rightarrow 0$, or when the diffusivity is large, $\mathcal{D} \rightarrow \infty$. In this limit, using l'Hôpital's rule, the solution reduces to

$$\hat{\phi} = \hat{x}, \quad (1.44)$$

which corresponds to a linear concentration profile associated with pure diffusion.

Figure 1.11 shows the non-dimensional concentration profile $\hat{\phi}$ as Pe increases from 0 to 100. As Pe increases, a region grows from the left in which advection dominates. This behavior arises because the coefficient $1/Pe$ in ODE (1.39) becomes small, causing the equation to reduce to

$$\frac{d\hat{\phi}}{d\hat{x}} \approx 0. \quad (1.45)$$

However, regardless of how large Pe becomes, a thin boundary layer always forms near $\hat{x} = 1$, where diffusion remains important so that the solution satisfies $\hat{\phi} = 1$ at $\hat{x} = 1$. Within this layer, the second derivative $d^2\hat{\phi}/d\hat{x}^2$ becomes large, so its product with $1/Pe$ in ODE (1.39) is not small.

1.6 The Burgers' equation

The advection–diffusion equation (1.5) is linear, in the sense that the velocity field $\mathbf{u}$ that transports ϕ is independent of ϕ itself. In contrast, the Navier–Stokes equations contain the nonlinear advection term $\mathbf{u} \cdot \nabla \mathbf{u}$, in which the velocity field advects itself. This nonlinearity can be studied in a simpler setting by dropping the pressure term from the Navier–Stokes equation, yielding Burgers' equation,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = \nu \nabla^2 \mathbf{u}. \quad (1.46)$$

By eliminating the pressure, Eq. (1.46) forms a closed system of three equations for the velocity components u , v , and w . This makes it easier to solve than the Navier–Stokes equation, which requires additional equations to determine the pressure p .

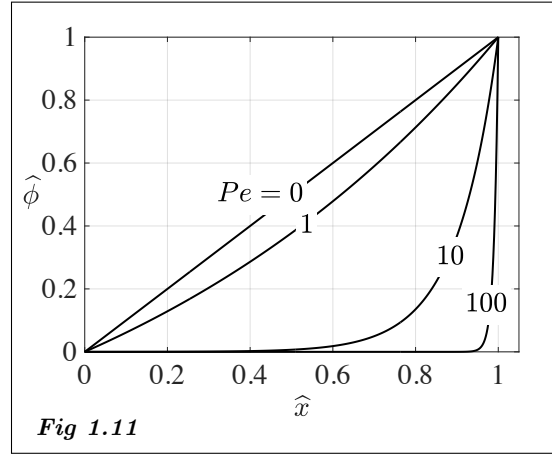


Fig 1.11

1.6.1 Nonlinear advection

Consider the one-dimensional Burgers' equation

$$\frac{\partial u}{\partial t} + \underbrace{u \frac{\partial u}{\partial x}}_{\text{nonlinear advection}} = \underbrace{\nu \frac{\partial^2 u}{\partial x^2}}_{\text{diffusion}}, \quad (1.47)$$

where $u = u(x, t)$. The major difference between Eq. (1.47) and the advection-diffusion equation is the nonlinear advection term

$$u \frac{\partial u}{\partial x}. \quad (1.48)$$

In the Navier–Stokes equation, this nonlinear term represents the advection of the x -component of momentum (ρu) by the velocity component u . To isolate the effects of nonlinear advection, we remove the diffusion term from Eq. (1.47) and consider

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0. \quad (1.49)$$

Suppose we solve Eq. (1.49) in the domain $0 \leq x \leq 1$, subject to periodic boundary conditions and the initial condition

$$u|_{t=0} = \sin(2\pi x), \quad (1.50)$$

which is shown as a red line in Fig. 1.12. In the region $0 < x < 0.5$, we expect this initial profile to be advected rightward by the positive velocity u . Meanwhile, in the region $0.5 < x < 1$, we expect the initial profile to be advected leftward.

With time, this causes $u(x, t)$ to become increasingly steep at $x = 0.5$, as illustrated by the dashed line in Fig. 1.12. Without the attenuating effects of diffusion, this eventually leads to the formation of a shock.

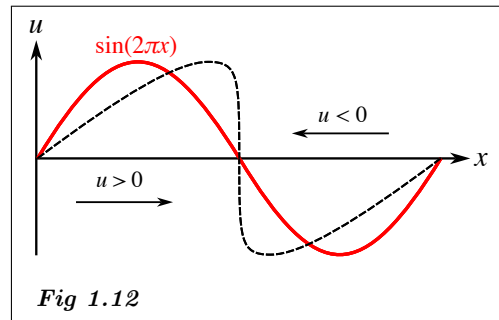
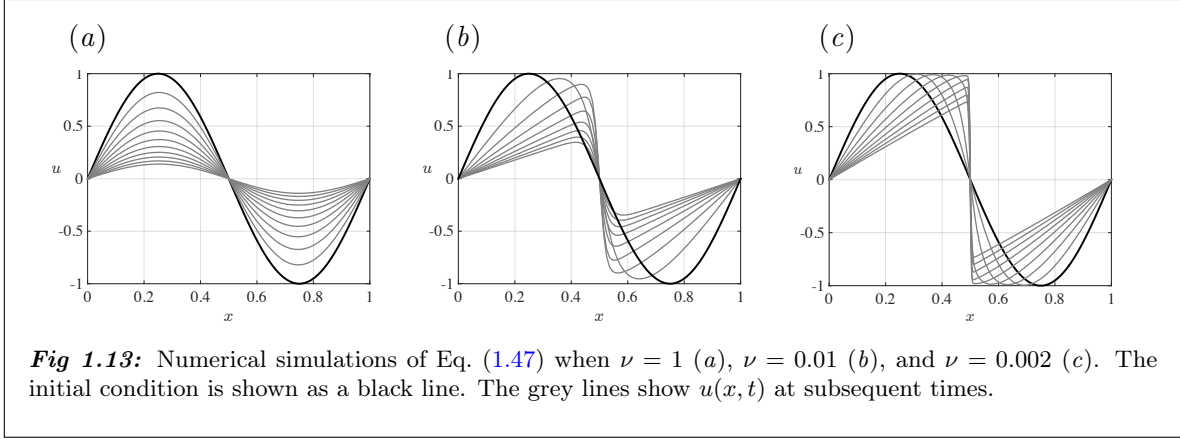


Fig 1.12



1.6.2 The viscous Burgers' equation

Let's now consider Burgers' equation (1.47), including the diffusion term. We solve Eq. (1.47) on the domain $0 \leq x \leq 1$ subject to the following boundary and initial conditions

$$u|_{x=0} = u|_{x=1} = 0, \quad u|_{t=0} = \sin(2\pi x). \quad (1.51)$$

We solve the problem numerically using a simple finite-difference method.

Figure 1.13(a) shows the resulting evolution of $u(x, t)$ when $\nu = 1$. For this large viscosity, diffusion dominates, so that $u(x, t)$ dissipates while maintaining its sinusoidal shape. Figure 1.13(b) shows the corresponding results when $\nu = 0.01$. In this case, diffusion reduces the peak value of $u(x, t)$, while advection causes the peak to shift toward $x = 0.5$. Finally, Fig. 1.13(c) shows the results when $\nu = 0.002$. For this small viscosity, advection initially dominates, moving the peak toward $x = 0.5$ without significant attenuation. Eventually, however, the gradient at $x = 0.5$ becomes large enough that diffusion begins to act, causing the late-time peak value of u to decrease.

1.7 Summary of key concepts

Key definitions:

1. **Advection:** the transport of a quantity by the bulk fluid motion.
2. **Diffusion:** the natural tendency of quantities such as mass, momentum, and energy to flow from regions of high to low concentration.
3. **Passive scalar:** a quantity, such as concentration, that is carried by the fluid without affecting its motion.
4. **Péclet number:** the nondimensional number representing a ratio of advection effects to diffusion effects. It is typically defined as $Pe = UL/\mathcal{D}$, where U and L are a characteristic velocity and length scale, and $\mathcal{D}$ is a diffusivity.

Key concepts:

1. Advection terms only propagate information downstream. In later chapters, we will see that this affects how we approximate these derivatives numerically.
2. Advection can create sharp gradients or discontinuities. These create challenges for CFD methods that assume smooth, differentiable fields.
3. Diffusion terms propagate information in all directions. They tend to smooth gradients rather than create sharp variations.
4. The relative importance of advection and diffusion can be expressed in terms of a nondimensional Péclet number. In later chapters, we will see that CFD often uses grid-based Péclet numbers based on the spacing between grid points.
5. Diffusion-dominated flows are usually easier to simulate because they are smooth. In contrast, advection-dominated flows typically need fine grids and special numerical treatments. This explains the variety of “advection schemes” in CFD software.